ITAI 1371 – Module 12

Reflective Journal Submission

Alexander DeBusk

Ethics, Fairness, and Bias in Machine Learning

L12 – Reflection – Alexander DeBusk

One of the biggest things I learned from this lab is that a machine learning model can have good overall accuracy and still be unfair to certain groups of people. Before this lab, I mostly thought of model performance in terms of how many predictions it got right. However, comparing the accuracy between males and females showed me why looking at the overall result is not always enough. The model had 81.20% accuracy for males and 91.81% for females, which is a noticeable difference.

Another important part of the lab was learning about False Positive Rates and False Negative Rates. I found it interesting that the model could make different types of mistakes depending on the group. Males had a higher False Positive Rate, while females had a higher False Negative Rate. This helped me understand that fairness is not just about getting the most predictions correct. It also matters who is being affected when the model makes mistakes.

I also learned that removing a sensitive feature, such as sex, does not automatically remove bias. Other information, like occupation, education, and working hours, can still be related to that feature. This is called proxy bias. It made me realize that even when a model does not directly use certain information, it can still learn patterns connected to it.

The most important challenge for me was understanding how to interpret the different fairness measurements and what they could mean in real situations. I can see why testing models across different groups is important before using them for decisions involving jobs, loans, or other opportunities.

ITAI 1371 – Module 12

Reflective Journal Submission

Shareefah Adeola

Ethics, Fairness, and Bias in Machine Learning

REFLECTION JOURNAL – Module 12 : Topic: Ethics, Fairness, and Bias in Machine Learning

ITAI 1371

Student: Shareefah Adeola

Date: November 12, 2026

This module provided valuable insight into the ethical challenges associated with machine learning, particularly how algorithms can inherit and reproduce existing societal biases. I learned about historical bias, measurement bias, and representation bias, and how these can affect the reliability and fairness of machine learning predictions.

During the practical exercise, the Adult Income dataset was used to train a Logistic Regression model. The model achieved an overall accuracy of 84.61%. However, when evaluated its performance separately for males and females, and discovered differences in their prediction accuracy. Female accuracy was 91.81%, compared to 81.20% for males.

Calculated the False Positive Rate (FPR) and False Negative Rate (FNR) for both groups. This exercise helped me understand that accuracy alone is not sufficient to determine whether a machine learning model is fair. A model may perform well overall while making different types of errors across demographic groups.

One important lesson I learned is that removing sensitive attributes such as gender does not necessarily eliminate bias because other variables may indirectly represent those attributes, known as proxy bias.

Overall, this module strengthened my understanding of responsible AI and the importance of fairness, transparency, and accountability in machine learning. Now I appreciate that developing a successful machine learning model involves more than achieving accurate predictions; it also requires ensuring that its outcomes are fair and do not disadvantage particular groups of people.

ITAI 1371 – Module 12

Reflective Journal Submission

Yilin Leng

Ethics, Fairness, and Bias in Machine Learning

Reflection

Before this lab, I mainly focused on overall accuracy when checking a machine learning model. I used to think that a model with high accuracy would perform well. However, this lab helped me understand that high overall accuracy does not always mean a model is fair to everyone. In this experiment, we used the UCI Adult dataset to train a Logistic Regression model and predict whether a person's yearly income is higher than $50,000. The model achieved an overall accuracy of 84.61%. However, when we checked the results for different genders, we found that the accuracy was 81.20% for males and 91.81% for females. This showed me that a model can perform differently for different groups, even when its overall accuracy looks good. Looking at the Confusion Matrix, I also learned more about the mistakes made by the model. For males, the False Positive Rate (FPR) was 10.26%, and the False Negative Rate (FNR) was 37.80%. For females, the FPR was 2.81% and the FNR was 47.84%. This means the model was more likely to incorrectly predict high income for males, while it was more likely to incorrectly predict low income for females. If this model were used for loan approvals or hiring, these mistakes could lead to unfair results. For example, some women with high incomes or good abilities might lose opportunities because of incorrect predictions. Another important thing I learned was the Proxy Bias. Removing a sensitive feature such as sex does not mean all bias will disappear. Other features, such as occupation, education, hours per week, and marital status, may still be related to gender. The model may still learn patterns related to gender through these features. Therefore, removing one feature is not enough to solve the problem. We also need to check the data and use methods such as reweighting and fairness constraints to reduce bias. Overall, this lab helped me understand that machine learning is not only about getting high accuracy. It is also important to

make sure that the model treats different groups fairly. Before using a model in the real world, we should check its performance for different groups and understand the mistakes it makes. In the future, I will pay more attention to data quality, fairness, and ethical issues when working with AI, instead of only trying to improve accuracy.